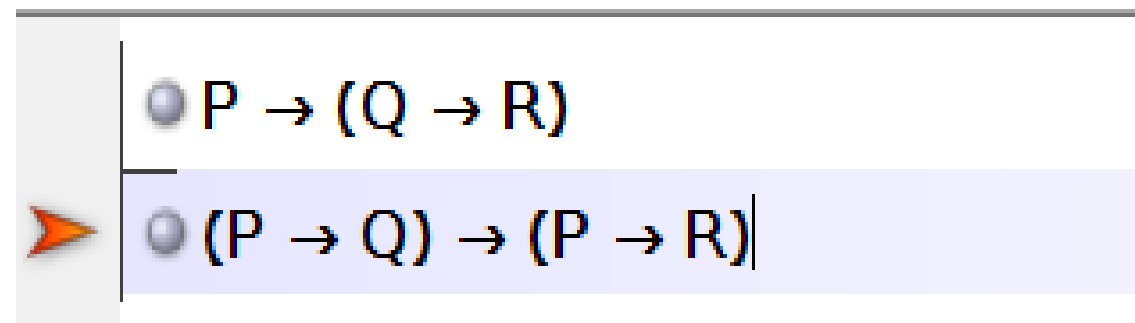
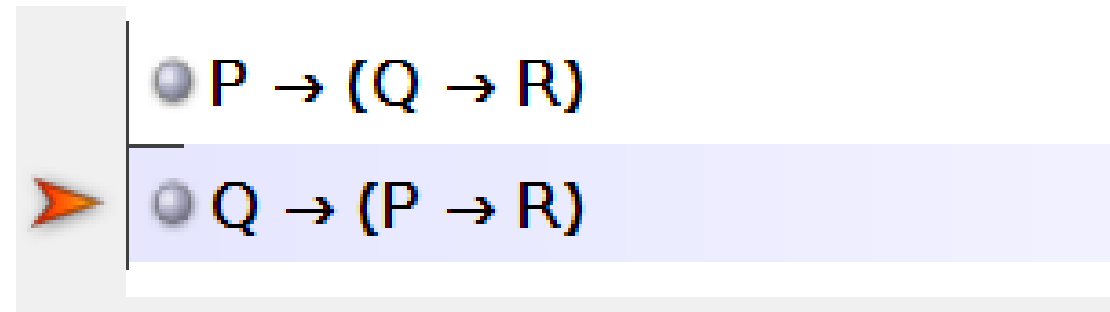
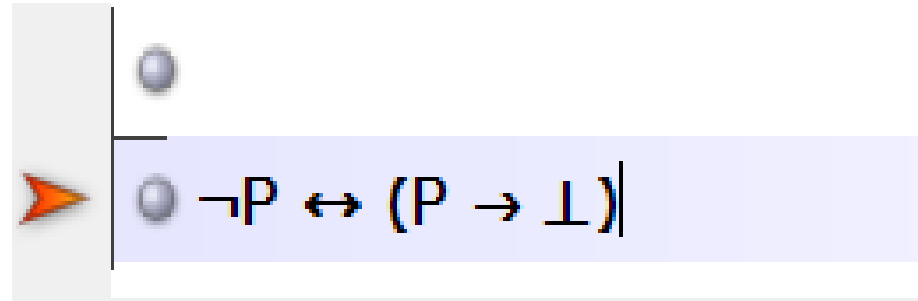


Esercitazioni

Fitch

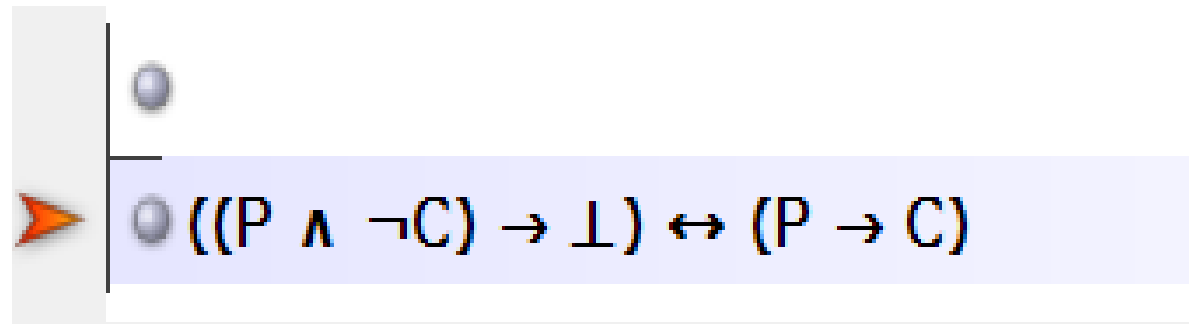


Fitch



A Fitch-style proof diagram. A vertical line on the left has a grey sphere at the top. A horizontal line extends from the vertical line. Below this line, the text $\neg P \leftrightarrow (P \rightarrow \perp)$ is displayed. An orange arrow points to the left of the text.

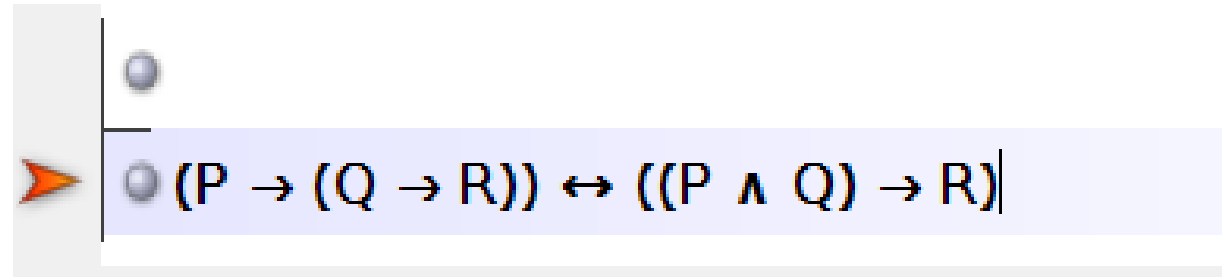
$$\neg P \leftrightarrow (P \rightarrow \perp)$$



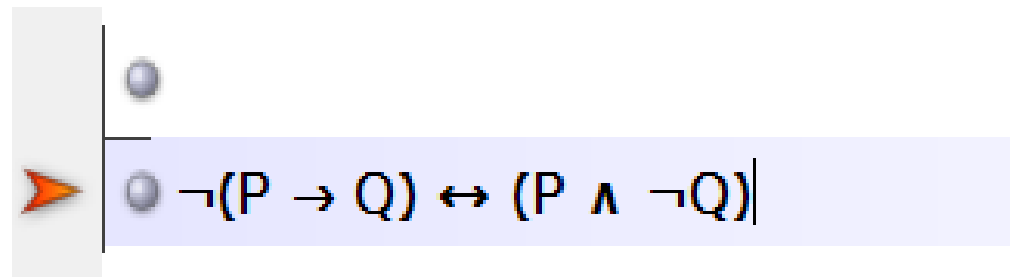
A Fitch-style proof diagram. A vertical line on the left has a grey sphere at the top. A horizontal line extends from the vertical line. Below this line, the text $((P \wedge \neg C) \rightarrow \perp) \leftrightarrow (P \rightarrow C)$ is displayed. An orange arrow points to the left of the text.

$$((P \wedge \neg C) \rightarrow \perp) \leftrightarrow (P \rightarrow C)$$

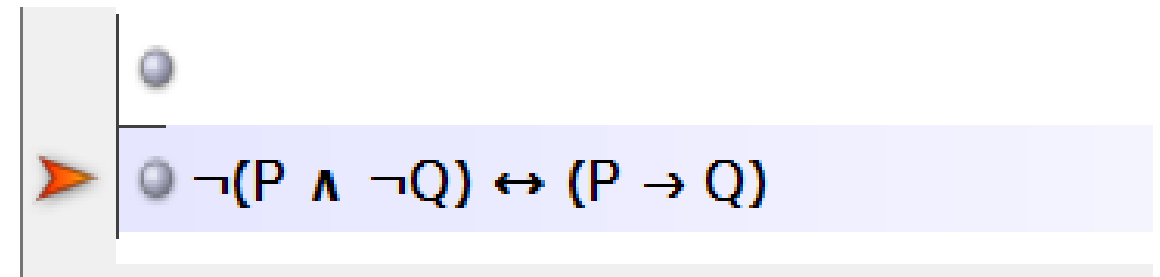
Fitch



$(P \rightarrow (Q \rightarrow R)) \leftrightarrow ((P \wedge Q) \rightarrow R)$



$\neg(P \rightarrow Q) \leftrightarrow (P \wedge \neg Q)$



$\neg(P \wedge \neg Q) \leftrightarrow (P \rightarrow Q)$

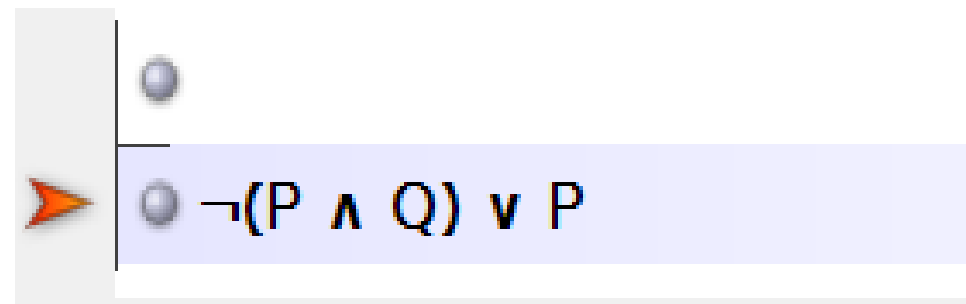
Fitch



●

● $P \vee \neg P$ ▼ Taut Con 🧐

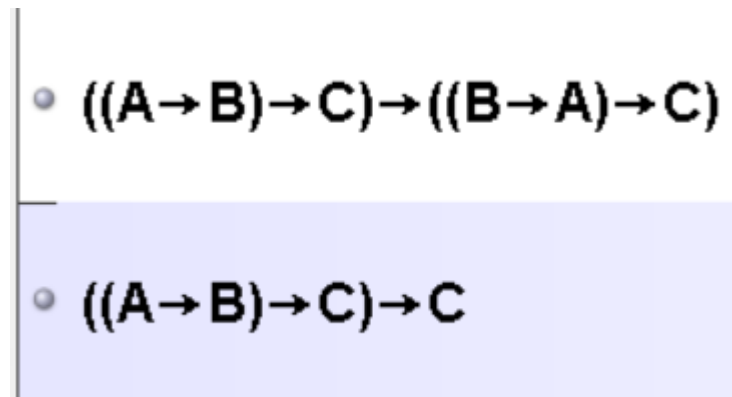
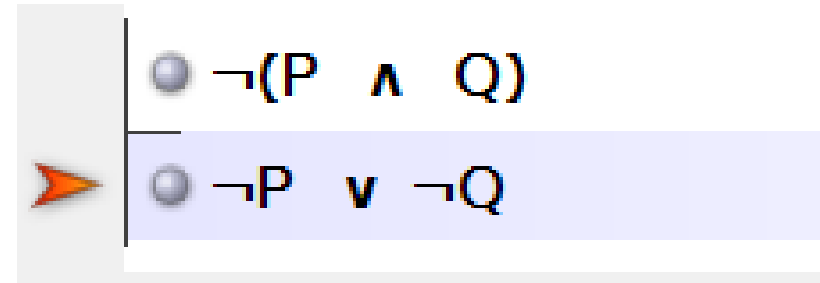
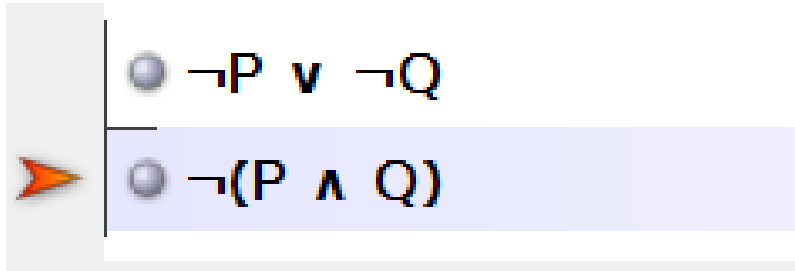
➤ ● $(P \rightarrow Q) \vee (Q \rightarrow P)$ ▼ Rule?



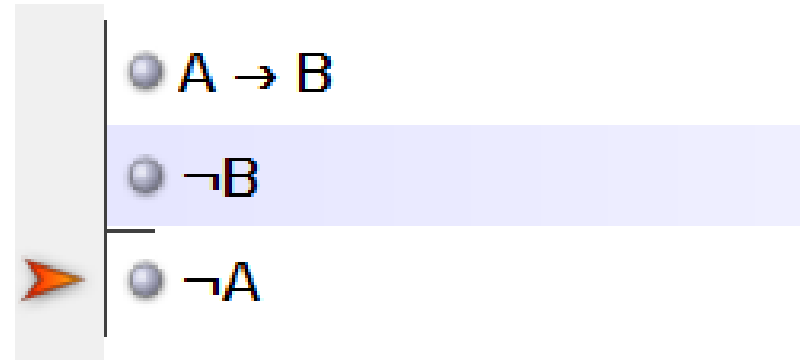
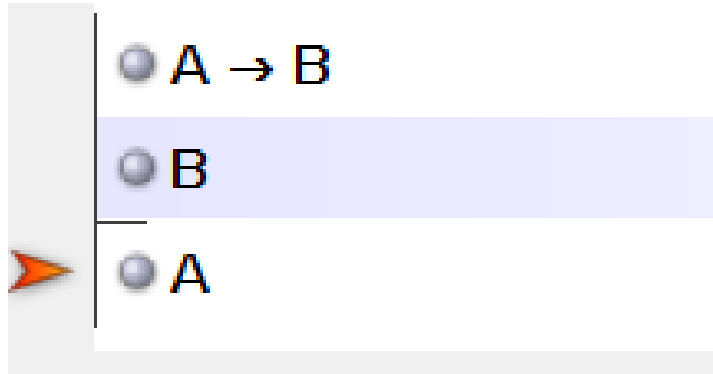
●

➤ ● $\neg(P \wedge Q) \vee P$

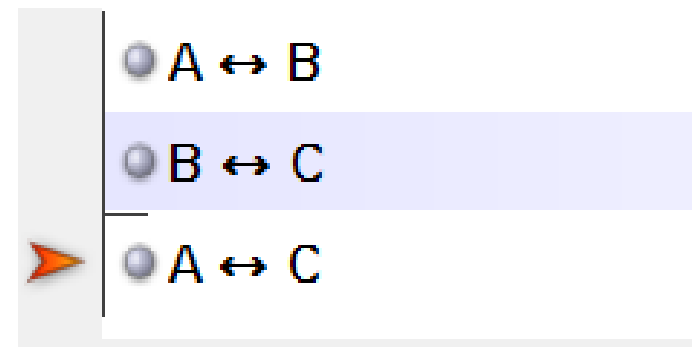
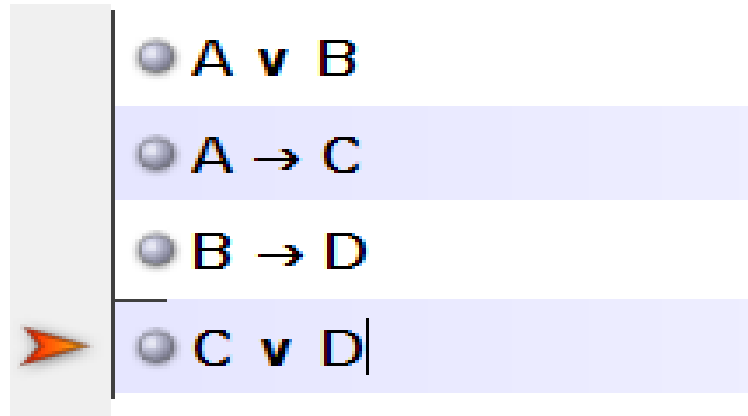
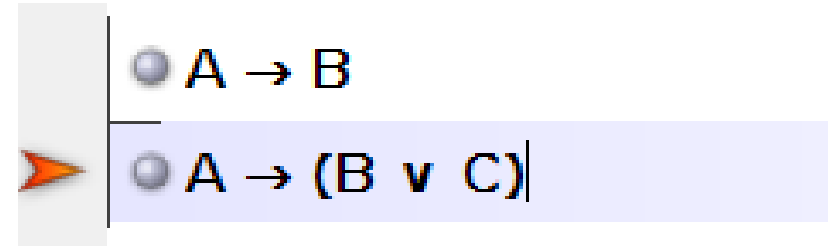
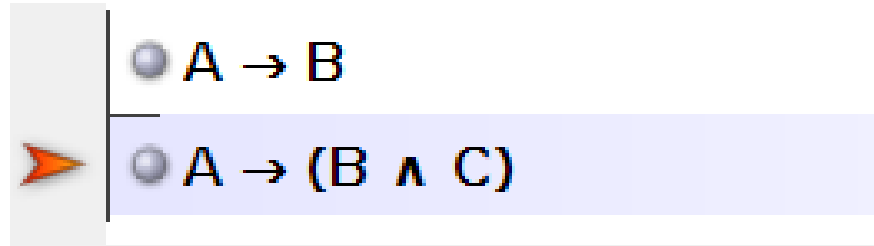
Fitch



Fitch(prova) o Tarski(controesempio)?



Fitch(prova) o Tarski(controesempio)?



Fitch(prova) o Tarski(controesempio)?

Ana Con permesso solo su atomiche, negate di atomiche e assurdo.

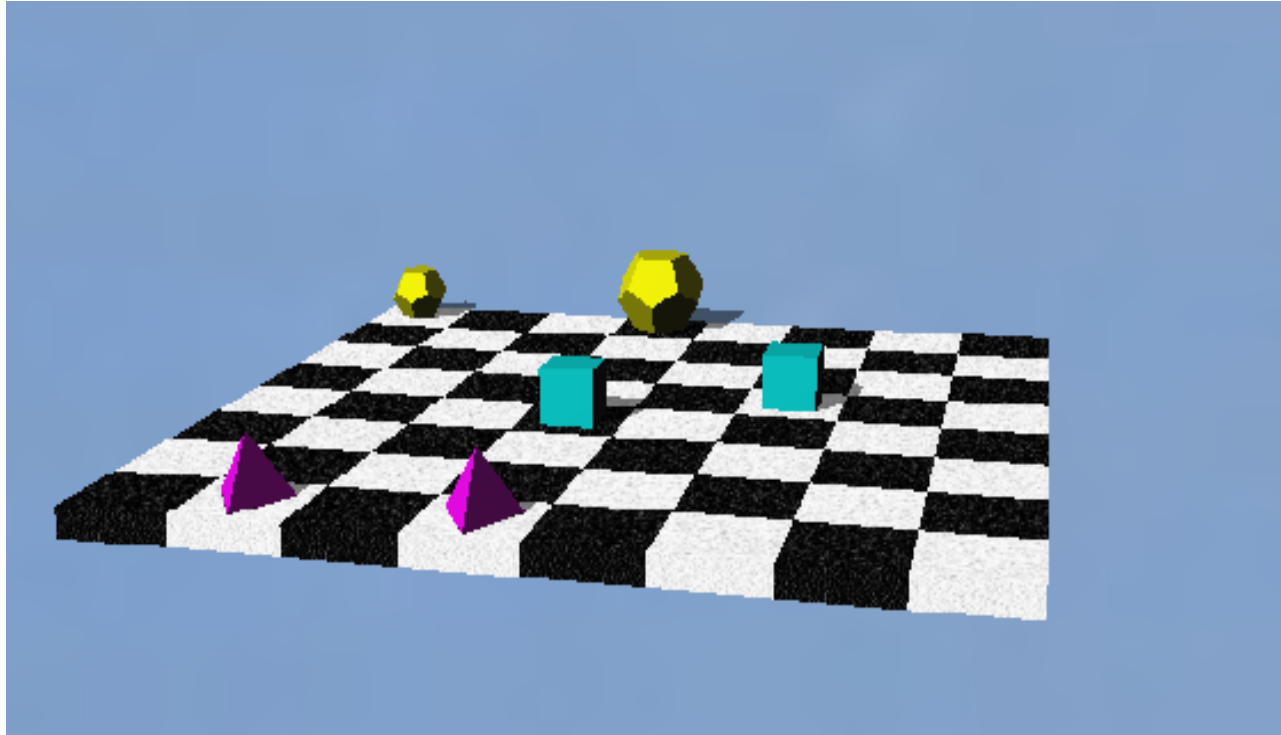
	Cube(c) $\vee$ Small(c)
	Dodec(c)
	Small(c)

	Larger(a, b) $\vee$ Larger(a, c)
	Smaller(b, a) $\vee$ $\neg$ Larger(a, c)
	Larger(a, b)

	$\neg(\neg\text{Cube}(a) \wedge \text{Cube}(b))$
	$\neg(\neg\text{Cube}(b) \vee \text{Cube}(c))$
	Cube(a)

	Dodec(b) $\vee$ Cube(b)
	Small(b) $\vee$ Medium(b)
	$\neg(\text{Small}(b) \wedge \text{Cube}(b))$
	Medium(b) $\wedge$ Dodec(b)

Tarski: dare nomi agli oggetti



1. $\text{Tet}(b) \leftrightarrow \text{Tet}(c)$

2. $\text{Dodec}(b) \leftrightarrow \text{Dodec}(c)$

3. $\text{Cube}(b) \leftrightarrow \text{Cube}(c)$

4. $\text{Tet}(a) \wedge \neg \text{Tet}(b)$

5. $\text{FrontOf}(a, b) \rightarrow (\text{FrontOf}(b, c) \vee \text{FrontOf}(c, b))$

6. $\text{LeftOf}(a, c) \rightarrow \neg \text{LeftOf}(a, b)$

7. $\text{BackOf}(b, a) \leftrightarrow \text{BackOf}(c, b)$

Traduzioni:

1. *If **a** is a tetrahedron, then **b** is also a tetrahedron.*
2. ***c** is a tetrahedron if **b** is.*
3. ***a** and **c** are both tetrahedra only if at least one of them is large.*
4. ***a** is a tetrahedron but **c** isn't large.*
5. *If **c** is small and **d** is a dodecahedron, then **d** is neither large nor small.*
6. ***c** is medium only if none of **d**, **e**, and **f** are cubes.*
7. ***d** is a small dodecahedron unless **a** is small.*
8. ***e** is large just in case it is a fact that **d** is large if and only if **f** is.*
9. ***d** and **e** are the same size.*
10. ***d** and **e** are the same shape.*
11. ***f** is either a cube or a dodecahedron, if it is large.*
12. ***c** is larger than **e** only if **b** is larger than **c**.*

Esercizi sulla sintassi

- Generare per strati le seguenti formule:

$$1. (A \wedge \neg B) \rightarrow (C \rightarrow (A \rightarrow B))$$

$$2. ((\text{Tet}(a) \vee \text{SameRow}(a,c)) \rightarrow \perp) \vee \text{Cube}(a)$$

- Quale delle seguenti stringhe non è una formula?

- $A \vee \neg \text{Tet}(a),$ $b = \text{socrate},$ $\text{Cube}(a \wedge b)$

Esercizi sulle equivalenze logiche

- Usando le equivalenze tautologiche notevoli che conoscete, trasformate la seguente formula:

$$(B \wedge ((C \vee (P \wedge \neg P)) \rightarrow C)) \vee \neg A$$

In una equivalente tale che:

1. Contenga solo il connettivo $\rightarrow$
2. Contenga il numero minimo di occorrenze di connettivi.

Domande:

- Applicare i Teoremi di Validità e Completezza per dire se le seguenti argomentazioni sono provabili o meno in **Fitch**.

- $A \wedge (B \vee \neg A \vee (C \wedge D))$
- $E \wedge (D \vee \neg(A \wedge (B \vee D)))$
- $A \wedge B$

- $A \wedge (B \vee \neg A \vee (C \wedge D)) \wedge \neg(A \wedge D)$
- $\neg(E \wedge (D \vee \neg(A \wedge (B \vee D))))$

Domande: definizioni

- Dare la definizione di tautologia
- Dare la definizione di enunciato (o formula) proposizionalmente insoddisfacibile
- Dare la definizione di equivalenza tautologica
- Dare la definizione di conseguenza tautologica

Domande: teoremi

- Enunciare il teorema di validità
- Enunciare il teorema di completezza
- Enunciare il teorema di deduzione

- Completare:

due enunciati P e Q sono tautologicamente equivalenti se e solo se l'enunciato è una tautologia.